Annex IV — Industrial Field Operation and Governed Synchronization of the WR Graph™

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex.

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, WR Graph™, Workflow Context Recorder (WCR), DeepFix sessions, BEAP™ transport (pBEAP™ and qBEAP™ classes), PoAC™-governed learning operations, and multi-device deployments comprising a central Host orchestrator and subordinate field devices or network workstations.

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

IV.0 Status, Scope, and Relationship to the Canon

This Annex specifies the industrial field operation of the WR Graph™: per-machine context regions, the taxonomy and governance of LoRA-class adapters in field deployments, the synchronization of WR Graph™ regions between a central Host and authorized field devices, internal (qBEAP™) session-template hosting, identity binding of internal handshakes, and multi-device session continuation.

This Annex does **not** redefine or extend wire formats, capsule serialization, cryptographic primitives, transport mechanisms, or PoAE™ execution semantics; these remain exclusively defined in the Canon and its Normative Annex. All enforcement mechanics of Annex I, the routing and gateway semantics of Annex II, and the acquisition governance of Annex III apply unchanged; nothing in this Annex weakens or contradicts them. Where this Annex refines the applicability of an existing rule to the intra-deployment case, the refinement is stated explicitly (see IV.5.8, IV.5.9).

Relationship to the Canon (v202). The behavior specified here was introduced in the Canon in the sections *"Optirando™ Station"*, *"Dynamic WR Code® Visualization & DeepFix Sessions"*, and *"From Fix to Knowledge: Building Verified Memories with WR Code®"*: per-machine WR Code® registration, DeepFix sessions over multimodal inputs, and the accumulation of a verified, semantically searchable per-machine **knowledge memory**. This Annex formalizes the mechanism behind that behavior. The terminology mapping in IV.1 is normative for interpretation: the concepts are continuous, and this Annex supersedes no conclusion of the Canon.

IV.1 Terminology Bridge to the Canon (Normative for Interpretation)

Canon (v202) term	This Annex / current Canon term
knowledge memory / knowledgebase (per WR Code®-registered asset)	WR Graph™ **context region** of that asset (IV.2)
allocated machine context, memory segments, historical logs	declarative and episodic **dimensions** of the asset's region
DeepFix session	DeepFix session (unchanged); its lifecycle is governed per IV.4
validated fix records with photos, logs, measurements	episodic evidence entities with provenance (Annex III §3)
Optirando™ Station (central deployment)	**Host** (IV.2.2)
adapter (deterministic, versioned integration module of the Canon)	**Integration Adapter** (unchanged; executable, capability-declared)
— (not present in v202)	**LoRA-class adapter** (parametric artifact; IV.3)

IV.1.1 Adapter Disambiguation (Normative). The Canon reserves the unqualified term *adapter* for Integration Adapters: deterministic, versioned, executable integration modules with formal capability declarations, executed within an isolated adapter execution host. The artifacts specified in IV.3 are **LoRA-class adapters**: non-executable parametric artifacts (tensor weights). The two classes SHALL NOT be conflated. Conforming documents SHALL use the qualified terms wherever ambiguity is possible.

IV.2 Machine Sessions as WR Graph™ Context Regions

IV.2.1 One Graph, Region-Structured (Normative)

A deployment SHALL maintain **one** authoritative WR Graph™. Individual machines, assets, or processes SHALL NOT be represented as separate, isolated graphs.

Each WR Code®-registered machine or asset anchors a **context region** of the deployment graph. A region comprises the four knowledge dimensions for that asset:

- **Declarative** — manuals, configurations, SOPs, known failure modes, manufacturer documentation
- **Episodic** — this asset's execution and maintenance history, DeepFix sessions, solved-issue records, feedback events, outcomes
- **Procedural** — approved repair and maintenance workflows (WCR recordings), checklists, automation paths
- **Parametric** — LoRA-class adapter bindings applicable to this asset (IV.3), per Annex II §II.2.1

A WR Code® scan resolves the asset's region and, via cross-dimensional edges, simultaneously selects what is known, what has happened, which parametric competence applies, and which automations may be offered (Annex II §II.2.4).

Rationale (Informative). A single region-structured graph preserves fleet-level capabilities the Canon already claims: recurrence statistics across identical machine models, DeepResearch-style pattern analysis across machine fleets, and transfer of validated fixes between assets of the same class. Per-machine graph islands would forfeit these by construction.

IV.2.2 The Host (Normative)

The **Host** is the deployment's primary orchestrator. The Host:

- SHALL hold the complete, authoritative WR Graph™ of the deployment, including all machine regions
- SHALL hold the deployment's internal session-template catalog (IV.6)
- SHALL be the exclusive point of graph mutation admission (IV.5.4); field devices contribute evidence but never mutate the authoritative graph directly
- MAY serve multiple concurrent machine sessions and multiple subordinate devices (field devices, network workstations)

Subordinate devices operate on **projections** (IV.5.3): grant-scoped subsets of the graph replicated for local, including offline, operation.

IV.2.3 Graceful Degradation (Normative)

Where a scan or query resolves no matching region — including assets not yet registered — the system SHALL degrade per Annex II §II.2.8: no adapter is loaded, no automation path is offered, no routing is fabricated, and the absence of domain knowledge for this asset is surfaced as such. A generic base-model answer MAY be given, clearly not attributed to asset-specific knowledge.

IV.3 LoRA-Class Adapters in Field Deployments

IV.3.1 Adapter Classes by Acquisition Path (Normative)

LoRA-class adapters in field deployments fall into exactly three classes, distinguished solely by acquisition path. All three are parametric artifacts in the identical technical sense; no class receives elevated authority.

1. **Field Adapters** — manufacturer- or OEM-provided domain adapters for specific device classes, machine families, or operational domains (e.g., a machine vision adapter for a specific PLC family, provided by its manufacturer).
2. **House-Internal Adapters** — adapters trained within the deployment on the organization's own material, including adapters trained on a specific machine's accumulated imagery, error states, and session evidence (IV.4).
3. **Optimization-Phase Adapters** — adapters produced by the Night Pipeline's data-set generation and training stage (Annex III §6.1, Stage 6) during the sorting and optimization phase of the WR Graph™.

House-Internal and Optimization-Phase Adapters differ only in initiating trigger (operator-initiated versus Night-Pipeline-proposed); their governance is identical.

The Source-entity acquisition enumeration of Annex III §3.3 is extended by the value **`manufacturer-provided`**. Field Adapters and their accompanying material SHALL be recorded as Sources with this acquisition value.

IV.3.2 Serialization and Integrity (Normative)

- LoRA-class adapters SHALL be distributed and stored exclusively in a non-executable tensor serialization format (safetensors or an equivalent format that structurally excludes embedded executable code). Serialization formats capable of code execution on load (e.g., pickle-based checkpoints) are prohibited.
- Every adapter artifact SHALL be integrity-bound by content hash. Hash verification SHALL occur at admission and at every load. Verification failure fails closed.
- Every adapter SHALL be bound to an identified base model, a declared purpose or task class, and a defined policy scope (per Canon M.5.3). An adapter presented against a mismatched base model SHALL NOT load.

IV.3.3 Authority Constraints — Suggest Mode (Normative)

LoRA-class adapters:

- carry **no operational authority**. They operate strictly in Suggest Mode: their outputs are hypotheses, never decisions or actions.
- SHALL NOT expand capability scopes, influence policy evaluation, or substitute for any PoAE™ authorization requirement (Annex II §II.2.3).
- SHALL NOT be treated as authoritative content. Ground truth is resolved from the declarative and episodic dimensions of the graph; adapter outputs SHALL be reconciled against graph content and presented as interpretation, not fact (per Canon M.5.2: capability states, not authoritative content).

Authority remains exclusively with: orchestration logic, the WR Graph™'s deterministic routing (preparatory, never executive — Annex II §II.2.5), the policy layer, PoAE™ enforcement, and the authenticated operator.

IV.3.4 Context-Scoped Activation (Normative)

- Adapter activation SHALL be triggered exclusively by deterministic context resolution (Annex II §II.2.2).
- **Field Adapters** (manufacturer-provided) SHALL be bound at admission to the context regions of the providing manufacturer's device classes and SHALL NOT activate outside those regions. A manufacturer's parametric competence applies exactly where that manufacturer is already trusted through its equipment, and nowhere else.
- Manufacturer identity SHALL be verified through DNS-based publisher verification, reusing the verification procedure required of BEAP™ Gateway operators (Annex II §II.3.2). No separate manufacturer PKI is introduced.

Trust rationale (Informative). A trusted manufacturer already shapes diagnostics through firmware, PLC software, HMI software, and the hardware itself. A manufacturer-provided adapter therefore shares the manufacturer's trust *anchor*. It does not share the equipment's trust *domain*: firmware runs on the machine and its faults manifest in machine behavior, whereas an adapter runs inside the operator's reasoning system and its faults manifest as skewed suggestions, which are harder to observe. Context-scoped activation, Suggest Mode, provenance display (IV.3.5), and telemetry (IV.3.6) are the compensating controls that make manufacturer provisioning safe without pretending the two trust domains are identical.

IV.3.5 Provenance Display (Normative)

Whenever an adapter contributes to a displayed suggestion, diagnosis, or interpretation, the user interface SHALL make the contributing adapter's identity and class (Field / House-Internal / Optimization-Phase) inspectable at the point of suggestion. Adapter influence SHALL NOT be silent.

IV.3.6 Telemetry, A/B Baseline, and Retirement (Normative)

Per-adapter telemetry, idle-time A/B comparison against the bare base model, and retirement proposals apply unchanged per Annex III §6.3. Field Adapters are not exempt: a manufacturer adapter that no longer outperforms the base model — including after a base-model upgrade — SHALL be proposed for retirement as a Class-3 mutation.

IV.3.7 Admission (Normative)

Admission of any LoRA-class adapter into the WR Graph™ — regardless of class — is a Class-3 mutation under PoAC™ (Annex III §4.4) and requires an explicit Consent Record, or coverage by an Organizational Authority Consent (IV.5.7) where deployment policy so provides. Until admitted, adapters remain in their libraries or in the sandbox trust domain and are unavailable to routing (Annex II §II.2.6).

IV.4 The Field Loop as a PoAC™ Instance

IV.4.1 Principle (Normative)

The industrial field loop — capture, evidence, gap, remediation, adapter — is an **instance of the PoAC™ acquisition discipline** (Annex III), not a parallel mechanism. No element of the loop mutates the WR Graph™ without an authorized, auditable path.

IV.4.2 Session Capture (Normative)

During a DeepFix session at a machine, the operator MAY contribute, and the system MAY capture:

- photographs (components, damaged parts, HMI screens, error codes, log screens)
- audio (verbal problem description) and its transcription
- free text
- video and live vision input (including smart-glasses capture)
- direct machine logs and telemetry via declared API connectors
- structural signals where a software surface is involved

All captured material enters as external-origin input through the Ingestor–Validator pipeline (Annex I §I.3), is recorded as Source entities and evidence events with full provenance (Annex III §§3.3, 5.2), and is scope-bound to the asset's region.

Media originals (photos, audio, video) SHALL be referenced from graph entities by content hash and handled as encrypted, inert blobs per the blob-inertness invariant; graph entities themselves remain structured textual artifacts (text-purity preserved).

IV.4.3 Session Feedback (Normative)

At session end, the operator's confirmation or rejection of the outcome SHALL be persisted as a Feedback Event per the schema of Annex III §5.2, with `target_ref` = the DeepFix session, `project_ref` = the asset's region, verdict, and optional commentary. This event is the primary evidence signal driving the asset's graph evolution.

IV.4.4 Evolution of the Region (Normative)

Accumulated evidence drives improvement through the Night Pipeline (Annex III §6) with the escalation ladder unchanged: workflow compilation → prompt profile → RAG enrichment → validator → tool integration → adapter training. In particular:

- Recurring fixes SHALL surface as process gaps, proposing WCR recordings of the validated repair procedure (Annex III §5.4, §5.7).
- Where evidence demonstrates that cheaper rungs cannot close a perception gap for an asset (e.g., visual identification of failure states specific to this machine), the Pipeline MAY propose **data-set generation from the asset's accumulated session imagery** and subsequent training of a House-Internal or Optimization-Phase adapter. Both steps are Class-3, consent-gated, with the data set as the persistent, provenance-bound asset and the adapter as its regenerable compilation (Annex III §3.4).
- Baseline checks apply: commodity knowledge already mastered by the base model SHALL NOT generate proposals (Annex III §5.3).

Operational effect (Informative). Over time, the asset's region converges on precise, efficient handling of recurring machinery issues. Because the region — not any individual — holds the validated fixes, the deployment retains diagnostic capability across shift changes and staff turnover: a technician on the next shift, or a newly onboarded one, scans the same WR Code® and inherits the full validated history, procedures, and trained perception for that machine.

IV.5 Graph Region Synchronization (Host ↔ Field)

IV.5.1 Two Context-Delivery Modes (Normative)

1. **Capsule-carried context (default, unchanged).** BEAP™ capsules carry the memory and context required for their task — global memory, agent memory, session context. This remains the general mechanism for both pBEAP™ and qBEAP™. No counterparty requires, and none SHALL receive, the WR Graph™ itself.
2. **Persistent region projection (internal-only, this Annex).** Within a single deployment's trust domain, an authorized device MAY hold a persistent projection of designated graph regions, maintained through synchronization roundtrips (IV.5.4).

IV.5.2 Transport Class Restriction (Normative)

- Graph region synchronization SHALL be carried exclusively over **qBEAP™** under an established internal handshake (IV.5.3, IV.7).
- A capsule carrying WR Graph™ structure or content over pBEAP™, or under a public or asymmetric handshake, is a protocol violation and SHALL fail closed at both endpoints.
- This rule is the internal-boundary counterpart of the cloud abstraction boundary (Annex III §7.3): WR Graph™ content crosses neither the cloud boundary nor the public-handshake boundary. Both edges of the same invariant.

IV.5.3 Sync Grants and Projections (Normative)

The internal handshake between Host and a subordinate device SHALL carry a **sync grant** (the internal application of the permission-grant model, Annex II §II.3.1), specifying:

- the granted **regions** (machines, assets, sites, or region groups)
- optionally, the granted **dimension depth** per region (e.g., full depth for network workstations; declarative + procedural + parametric with condensed episodic history for a field tablet)
- validity period and revocation conditions

Projection SHALL be derived exclusively from the sync grant. Content outside the grant SHALL NOT be transmitted; a request outside the grant SHALL fail closed. Default is deny.

IV.5.4 Roundtrip Model (Normative)

A synchronization roundtrip comprises three stages, reusing the versioned, hash-addressable context-block model of the Relational Context Graph (Annex I, *Local Persistence and Incremental Enrichment*):

1. **Inventory.** The device transmits its per-region version cursor or block-hash inventory for granted regions.
2. **Delta (down-sync — replication).** The Host responds with graph-sync capsules containing only the missing or updated blocks for granted regions, at granted depth. Down-sync is replication of already-admitted graph content; it is not a new admission.
3. **Contribution (up-sync — evidence).** The device transmits session output as ordinary evidence capsules: episodic events, captured media references, logs, and Feedback Events from DeepFix sessions.

Deltas follow the append-and-enrich model: blocks are created once, referenced by `(block_id, hash)`, and enriched incrementally; full-payload retransmission is avoided by design.

IV.5.5 Host Authority and the Absence of Merge (Normative)

- The Host's graph is authoritative. Subordinate devices SHALL NOT mutate the authoritative graph directly.
- Up-synced contributions SHALL traverse the Host's Ingestor–Validator pipeline (Annex I §I.3) and are admitted under PoAC™ classification like any other material.
- There is no bidirectional graph merge. Conflicting field observations are not a merge conflict: they are distinct episodic events, both retained with provenance and temporal

ordering. Contradiction handling follows bi-temporal invalidation (Annex III §3.7), performed on the Host.

Security effect (Informative). A compromised or lost field device cannot poison the authoritative graph directly; at worst it submits malformed or malicious evidence, which is precisely the input class the ingestion pipeline exists to contain.

IV.5.6 Payload Classes (Normative)

- **Graph deltas** are structured textual artifacts (text-purity preserved).
- **Media blobs** referenced by synced entities travel as encrypted attachments, hash-bound, and remain inert on every device (blob-inertness preserved). Devices MAY receive blobs lazily (on demand) within their grant.
- **Adapter weights** (parametric dimension) MAY be projected to devices whose grant covers the corresponding regions, so that offline field diagnosis with the asset's trained perception is possible. Weights sync under IV.3.2 integrity rules. Intra-deployment replication under an internal handshake occurs **within** the user's trust domain in the sense of Annex II §II.2.3; the device inherits the Host's admission consent (IV.5.7) and SHALL NOT require re-consent per device.

IV.5.7 PoAC™ Classification of Synchronization (Normative)

- **Down-sync** of already-admitted content to a granted device does not constitute a new Class-3 acquisition. It SHALL be covered by an **Organizational Authority Consent**: a persisted Consent Record, issued under the deployment's policy authority (the policy-gate mechanism of Annex III §3.6), that covers a declared class of synchronized content for declared device classes. Issuance, modification, and revocation of an Organizational Authority Consent are themselves Class-3 policy mutations and are logged accordingly.
- **Up-sync** contributions classify normally under Annex III §4: new Sources and evidence are Class-3 or covered by a standing Organizational Authority Consent for the declared evidence class; Class-1 linking and condensation over admitted events proceeds per the assignment table.
- Nothing in this section weakens §4.4: no content enters the authoritative graph without a Consent Record; this section defines *which* Consent Record covers replication and routine field evidence, so that per-delta operator consent across a device fleet is not required.

IV.5.8 Intra-Deployment Evidence and Routing State (Normative Refinement of Annex II §II.2.5)

Annex II §II.2.5 excludes external signals — including incoming capsules — from routing state. For the purposes of that rule, devices operating under the same organizational authority, over internal (qBEAP™) handshakes established per this Annex, are **not external parties**: their up-synced evidence constitutes local, operator-authorized signal once admitted through the Host's ingestion pipeline. Evidence arriving over public or asymmetric handshakes, from publishers, or from any party outside the deployment's organizational authority remains external and SHALL NOT influence routing state, unchanged.

IV.5.9 Revocation and Projections at Rest (Normative)

- Revocation of a device's internal handshake SHALL immediately prevent all future synchronization roundtrips under it (consistent with Annex I §I.4.2).
- Projections at rest on the device are protected by the custody invariant of IV.7.4: projection storage keys derive from the authenticated operator session; without a valid, second-factor-authenticated session, the projection is cryptographically inert.
- In High-Assurance Mode (Annex I §I.8), deployments SHALL additionally support a **wipe-on-next-contact** policy: a revoked device that reconnects receives a destruction directive for its projection before any other exchange, and the wipe result is logged.

IV.6 Session-Template Hosting: Public and Internal Classes

IV.6.1 Two Template Classes (Normative)

1. **Public templates (pBEAP™).** Session templates referenced by publicly scannable WR Code®s. They are hosted in the public Optirando™ repository (optirando.com), are publicly inspectable, and are exchanged under asymmetric, post-quantum-ready end-to-end encrypted handshakes. This class is defined in the Canon and is unchanged by this Annex.
2. **Internal templates (qBEAP™).** Session templates referenced by internal WR Code®s, scannable only by enrolled users of the deployment. They are hosted on and loaded from the **Host**, delivered over qBEAP™ under internal handshakes. They are never published to, and never resolvable through, the public repository.

The template taxonomy (structure, capability declarations, stamping) is identical across both classes; the classes differ exclusively in visibility, hosting locus, and handshake class.

IV.6.2 Internal Template Governance (Normative)

Internal templates SHALL carry governance metadata binding at minimum: version, author identity, **approval state**, capability references, integrity verification (WRStamp), and **organizational scope**. Version, author, capabilities, integrity, and lineage are already bound by stamped code capsules (Annex I §I.7.2); approval state and organizational scope are the additive fields.

- **Organizational scope** SHALL restrict resolution and execution to devices enrolled in the deployment (holding a valid internal handshake and, where required by policy, attestation). An internal WR Code® scanned by a non-enrolled device SHALL resolve to nothing; it SHALL NOT fall back to public resolution.
- Admission of an internal template into the Host catalog, and every revision, is a Class-3 mutation under PoAC™ (Annex III §4.4), subject to the WCR authoring and validation lifecycle where the template embeds recordings (Annex II §II.5). Background-proposed revisions follow Annex II §II.4: proposed, never applied silently.
- Organizational scope is recipient-binding at group granularity. It composes with, and never weakens, per-capsule recipient binding.

IV.6.3 Resolution Rule (Normative)

WR Code® resolution SHALL determine template class deterministically from the code's registration: public codes resolve against the public repository over pBEAP™; internal codes resolve against the Host over qBEAP™. No code SHALL be resolvable through both paths.

IV.7 Identity Binding and Cryptographic Hardening of Internal Handshakes

IV.7.1 Layered Identity Model (Normative)

Internal deployments operate three identity layers. The layers are complementary; none substitutes for another.

1. **Endpoint identity** (inter-party, unchanged): anchored to a confirmed email address; enterprise-anchored endpoint identities additionally require DNS-verified domain control and confirmed organizational address ownership (Canon A.3.054.3). This layer provides **identity proofing at establishment** — it determines *with whom* a handshake is concluded. It provides no runtime protection of handshake material.
2. **Operator identity** (intra-deployment): the operator's SSO identity at the deployment's identity provider — the stable **subject identifier and realm** — with **mandatory second-factor enforcement** at session login to the orchestrator.
3. **Device identity**: the device's registered asymmetric key material (orchestrator device keys), bound at enrollment. In High-Assurance Mode, device identity keys are attestation-bound (IV.7.5).

IV.7.2 Threat Statement (Informative)

Without the mechanisms of IV.7.3–IV.7.5, an established handshake degenerates into a pure possession credential: whoever obtains the handshake material can exercise the relationship, and the establishment-time email anchoring contributes nothing at the moment of use. This Annex therefore treats endpoint anchoring as necessary but insufficient, and adds custody protection (at rest), derivation binding (in use), and attestation binding (environment integrity) as independent, cumulative hardening layers.

IV.7.3 Cryptographic Derivation Binding (Normative)

- The handshake identifier and all handshake-derived eligibility material for internal handshakes SHALL be derived over, at minimum: the sender and receiver endpoint identities (per Canon A.3.054.3), the **operator identity** (stable subject identifier and realm), the **device identity** (device public key), and the handshake context.
- Verification at either endpoint SHALL fail closed where the presenting party cannot prove the bound operator identity and device identity. Consequently, handshake material exercised from a different operator identity, or from a different device, fails cryptographic verification at the counterparty — not merely a local policy check.

- Binding SHALL be to **stable identifiers only** (subject identifier, realm, device public key) — never to session tokens, access tokens, or other rotating session artifacts. Rotating artifacts gate *use* (IV.7.4); stable identifiers define *whose* handshake it is.
- Sync grants (IV.5.3) and Organizational Authority Consents (IV.5.7) SHALL be bound to the same identifiers.

IV.7.4 Session Gating and Custody Invariant (Normative)

- **Use** of a bound handshake or grant SHALL require a live, second-factor-authenticated operator session at the deployment's identity provider. Second-factor enforcement SHALL be mandatory at the identity provider (a required authentication step, not a user-optional setting).
- Handshake material, sync-grant records, and projection storage keys on any device SHALL reside in vault storage whose keys derive from the authenticated operator session. Consequently: possession of a device, or exfiltration of its storage (including the handshake ledger), does not yield usable handshake material absent a valid second-factor-authenticated session; and logout renders handshake material and projections cryptographically inert on that device until the next authenticated login.

IV.7.5 Attestation Binding (High-Assurance, Normative)

In High-Assurance Mode (Annex I §I.8):

- Device identity keys SHALL be generated within, and certified by, a hardware-attested execution environment (e.g., TPM- or TEE-resident keys with attestation evidence), such that the device identity itself carries proof of environment integrity.
- Handshake establishment SHALL verify the attestation evidence of the device identity key; the attested key enters the handshake derivation per IV.7.3, transitively binding the handshake to environment integrity. Material presented from a non-attested or tampered environment fails cryptographically.
- Attestation SHALL be re-validated per deployment policy (on interval or on demand, consistent with Canon A.3.054.4's dynamic verifiability). Re-validation failure SHALL suspend use of all handshakes bound to the affected device identity until attestation is restored.

IV.7.6 Coverage and Residual Risk (Informative)

Each layer defeats a distinct attack class: endpoint anchoring defeats establishment-time impersonation; the custody invariant defeats offline theft and loss of logged-out devices; derivation binding defeats replay of exfiltrated material under a different identity or from a different device, including insider misuse of another operator's material; attestation binding defeats use from integrity-compromised environments. **Not covered by identity binding:** an adversary executing on a device during a live authenticated session acts with the operator's identity. This residual class is addressed by the enforcement architecture, not by handshake identity: per-step PoAE™ gating of consequential actions, the mandatory sandbox model (Annex I §I.2), and attestation re-validation.

IV.7.7 Enterprise Designation Compatibility (Normative)

Where a deployment operates Enterprise Handshakes (Canon A.3.054.4: hardware-attested orchestrators, enterprise-anchored endpoint identities), the requirements of this section are additive. Nothing in this Annex relaxes attestation or endpoint-anchoring requirements.

IV.7.8 Revocation Path (Normative)

Handshakes established under this Annex MAY be long-lived, including without scheduled expiry. Absence of expiry SHALL NOT mean absence of revocation. The following SHALL each independently and immediately terminate usability of a handshake and all grants under it:

1. explicit handshake revocation at the Host (per Annex I §I.4.2 semantics: future activation denied, historical integrity preserved);
2. disablement or deletion of the operator identity at the identity provider — no authenticated session can be established, vault keys become underivable, and handshake material is rendered inert (IV.7.4); additionally, derivation binding (IV.7.3) invalidates any attempt to re-bind the material to a substitute identity;
3. revocation of the device's enrollment (device identity), with wipe-on-next-contact where High-Assurance policy applies (IV.5.9).

IV.8 Session Continuation and Multi-Device Operation

IV.8.1 Session State as Synced Content (Normative)

DeepFix session state — captured inputs, intermediate findings, adapter suggestions with provenance, pending confirmations — SHALL be representable as region-scoped content subject to the synchronization model of IV.5. A session opened on one device MAY, within the operator's grant, be continued on another enrolled device after a sync roundtrip.

IV.8.2 Quick and Deep Sessions (Informative)

A smartphone or field tablet suffices for quick sessions at the machine: scan, capture, receive suggestions, confirm, rate. For complex machinery, the operator continues the same session at a multi-screen workstation — full graph depth, side-by-side comparison of history and hypotheses, full documentation review. The continuation is the same session against the same region; no re-capture and no re-entry of context occurs. This mirrors, and formalizes, the Canon's capture-in-the-field / process-after-sync pattern.

IV.8.3 Consequential Steps (Normative)

Session continuation never carries authorization across devices implicitly. Every consequential step remains gated at its applicable PoAE™ tier on the device where it is confirmed; a confirmation pending on device A is not satisfiable by mere session arrival on device B.

IV.9 Worked Example — Field Technician at a Registered Machine (Informative)

A technician on the late shift finds packaging machine M-747 halting intermittently. She scans its WR Code® with a field tablet.

Resolution. The scan resolves M-747's context region. The graph selects across dimensions simultaneously: declarative (the machine's manual, its configuration, known failure modes for this model), episodic (M-747's own history — the same symptom occurred twice, four months apart), procedural (the approved inspection workflow recorded after the last occurrence), parametric (the House-Internal adapter trained, with consent, on M-747's accumulated session imagery).

Capture. She photographs the HMI error screen, the drive assembly, and a worn coupling; dictates a description; pulls the last 200 controller log lines over the declared API connector. Everything enters through ingestion as provenance-bound evidence, scoped to M-747's region.

Interpretation. The adapter — identified as such on screen, with its class and lineage inspectable — flags the coupling wear pattern as matching the failure state from the prior episodes. The graph reconciles: episodic records show the previous validated fix, including the photos the day-shift technician attached back then. The system proposes the recorded repair workflow.

Governed execution. The workflow runs to the authorization boundary. Ordering the replacement part is a consequential step; it halts at its PoAE™ tier until she confirms. Nothing the adapter suggested, and nothing the routing surfaced, pre-satisfied that confirmation.

Feedback. The fix works. She confirms; a Feedback Event with positive verdict and a short comment is persisted against the session and M-747's region.

Evolution. Overnight, the Night Pipeline links the event, updates confidence on the validated fix, and notes the recurrence. Because this failure has now occurred three times, it proposes (Class 3, in the Morning Digest, with the recurrence evidence attached) a preventive-maintenance workflow recording. The maintenance lead accepts; the procedure enters the region with full provenance.

Continuity. A week later, a newly hired technician on a different shift scans the same code. He has never seen this machine. He inherits everything: history, validated fixes, the trained perception, the new preventive procedure. If he scans an unregistered machine instead, the system says so — it offers no fabricated confidence, loads no adapter, and answers only from the base model, clearly marked as non-asset-specific.

IV.10 Relation to Prior Work and Claimed Contribution (Informative)

The constituent mechanisms — per-asset knowledge bases, maintenance-record retrieval, edge anomaly detection, LoRA adaptation, dataset-driven fine-tuning, device synchronization — exist independently in prior work and in the Canon itself. The contribution claimed by this Annex is the **normative composition**: per-asset context regions of a four-dimensional graph, whose evidence-driven evolution (including the training of asset-specific parametric competence from field-captured material) is governed end-to-end by PoAC™ acquisition discipline, whose replication to field devices rides exclusively on the deployment's existing qBEAP™ enforcement stack under grant-scoped projection with host-exclusive admission authority, and whose every suggestion remains subordinate to PoAE™ execution gating. Capability acquisition, capability distribution, and capability exercise are placed under one continuous proof discipline.

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§Version History

Version	Date	Change
1.0	2026-07-08	Initial normative specification: terminology bridge to Canon v202 (knowledge memory → WR Graph™ context regions), region model and Host definition, LoRA-class adapter taxonomy (Field / House-Internal / Optimization-Phase) with serialization, authority, activation-scoping, provenance-display and telemetry rules, field loop as PoAC™ instance, qBEAP™-exclusive graph region synchronization with sync grants, host-authoritative contribution model and Organizational Authority Consent, internal (qBEAP™) template hosting with organizational scope, layered identity binding with custody invariant and revocation paths, session continuation, worked example.

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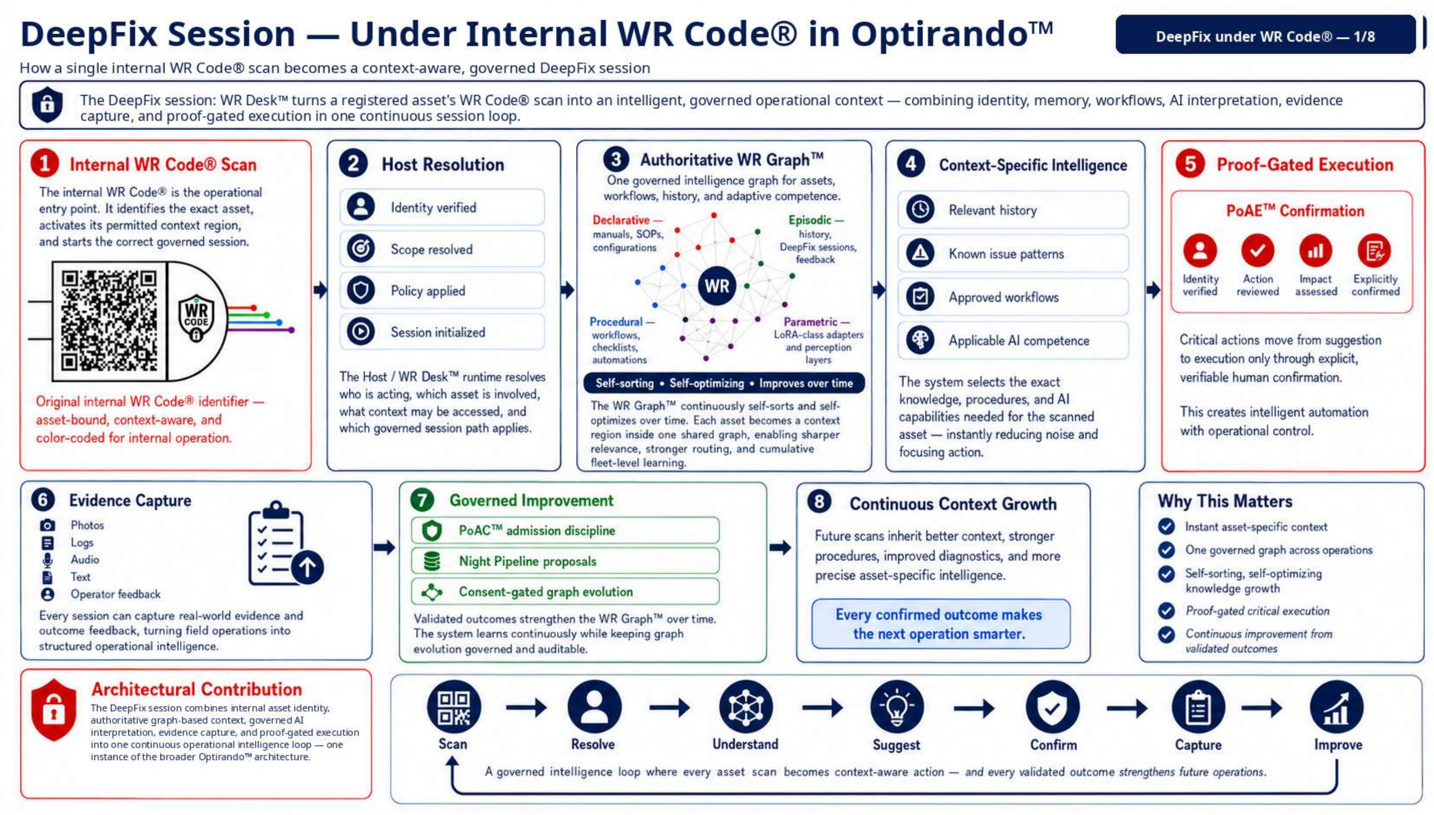
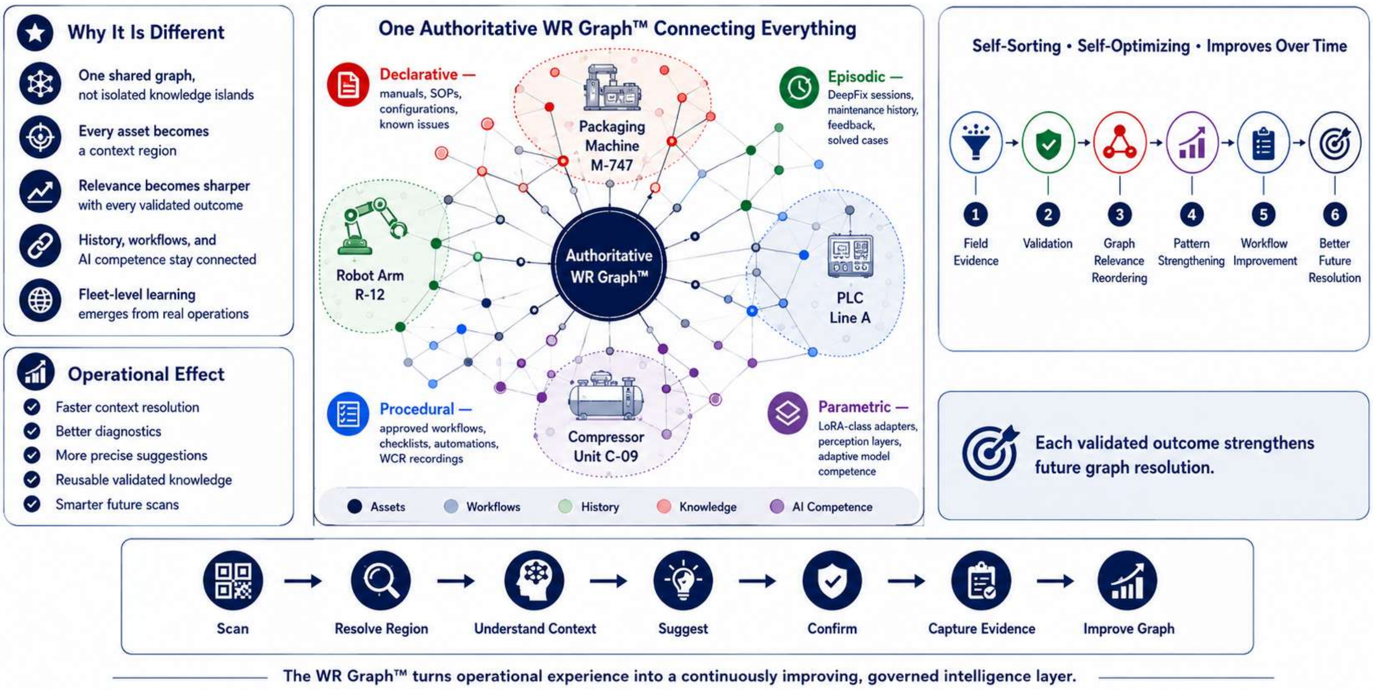


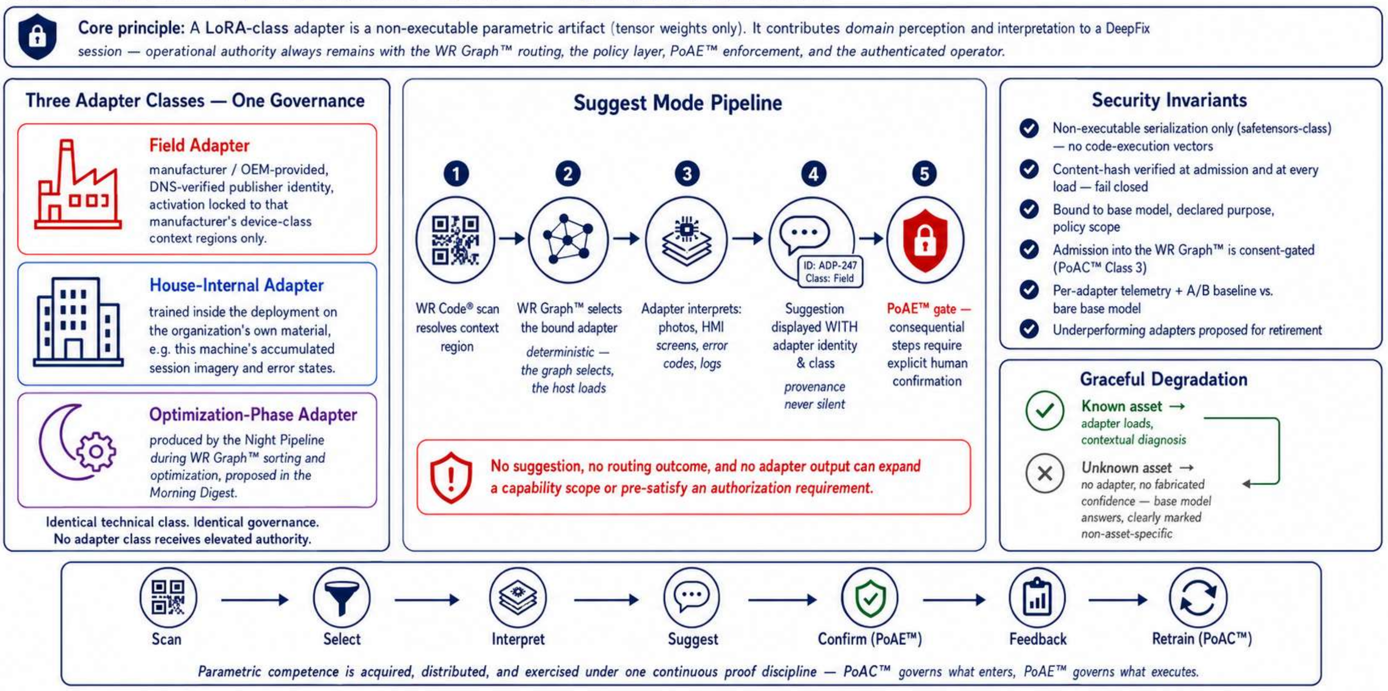
WR Graph™ — Self-Sorting, Self-Optimizing Intelligence Graph

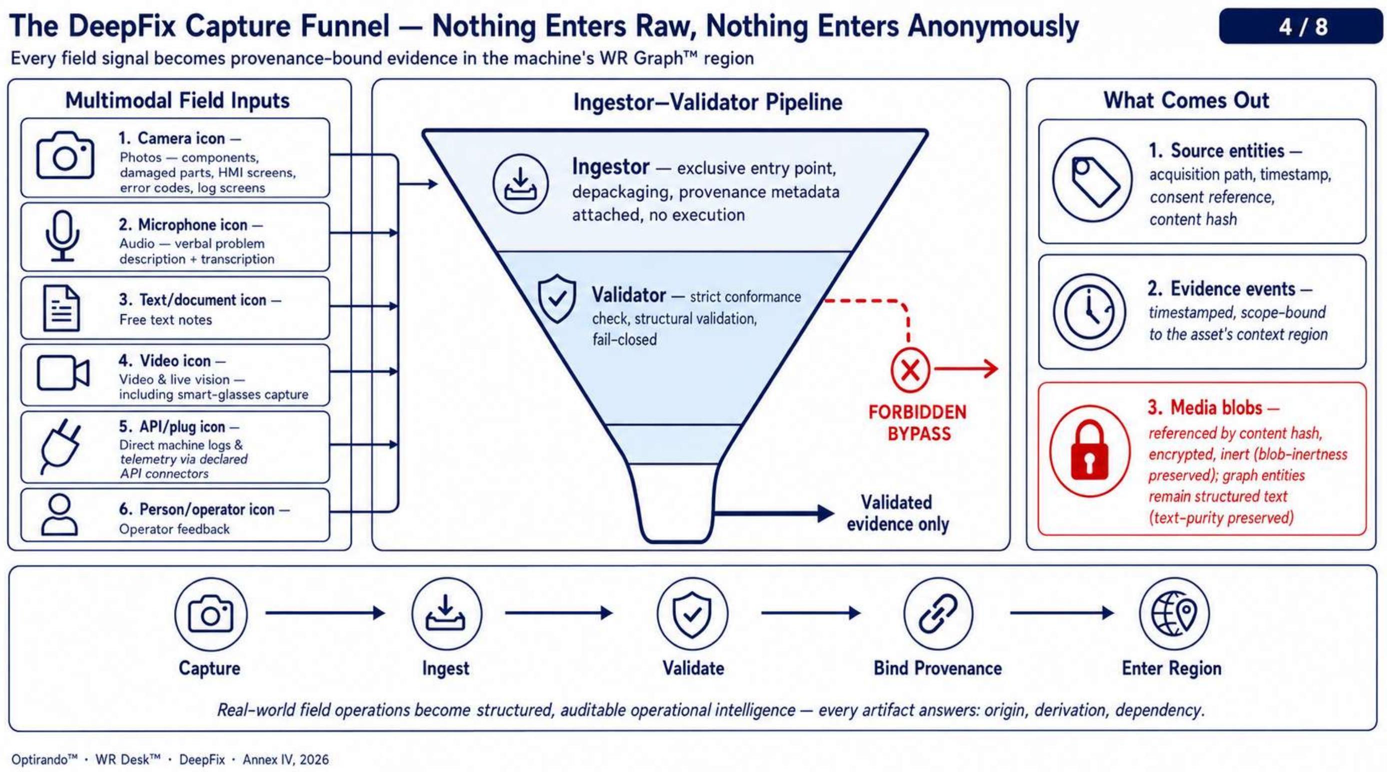
Infographic 2 of 8 • The context layer behind every DeepFix session — one authoritative graph for assets, workflows, history, and adaptive intelligence



LoRA-Class Adapters — Competence Without Authority

How DeepFix gains machine-specific perception under Optirando™ — adapters propose, they never decide




The DeepFix Capture Funnel — Nothing Enters Raw, Nothing Enters Anonymously
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Every field signal becomes provenance-bound evidence in the machine's WR Graph™ region
Multimodal Field Inputs
1. Camera icon — Photos — components, damaged parts, HMI screens, error codes, log screens
2. Microphone icon — Audio — verbal problem description + transcription
3. Text/document icon — Free text notes
4. Video icon — Video & live vision — including smart-glasses capture
5. API/plug icon — Direct machine logs & telemetry via declared API connectors
6. Person/operator icon — Operator feedback
Ingestor–Validator Pipeline
Ingestor — exclusive entry point, depackaging, provenance metadata attached, no execution
Validator — strict conformance check, structural validation, fail-closed
FORBIDDEN BYPASS
Validated evidence only
What Comes Out
1. Source entities — acquisition path, timestamp, consent reference, content hash
2. Evidence events — timestamped, scope-bound to the asset's context region
3. Media blobs — referenced by content hash, encrypted, inert (blob-inertness preserved); graph entities remain structured text (text-purity preserved)
Capture
Ingest
Validate
Bind Provenance
Enter Region
Real-world field operations become structured, auditable operational intelligence — every artifact answers: origin, derivation, dependency.
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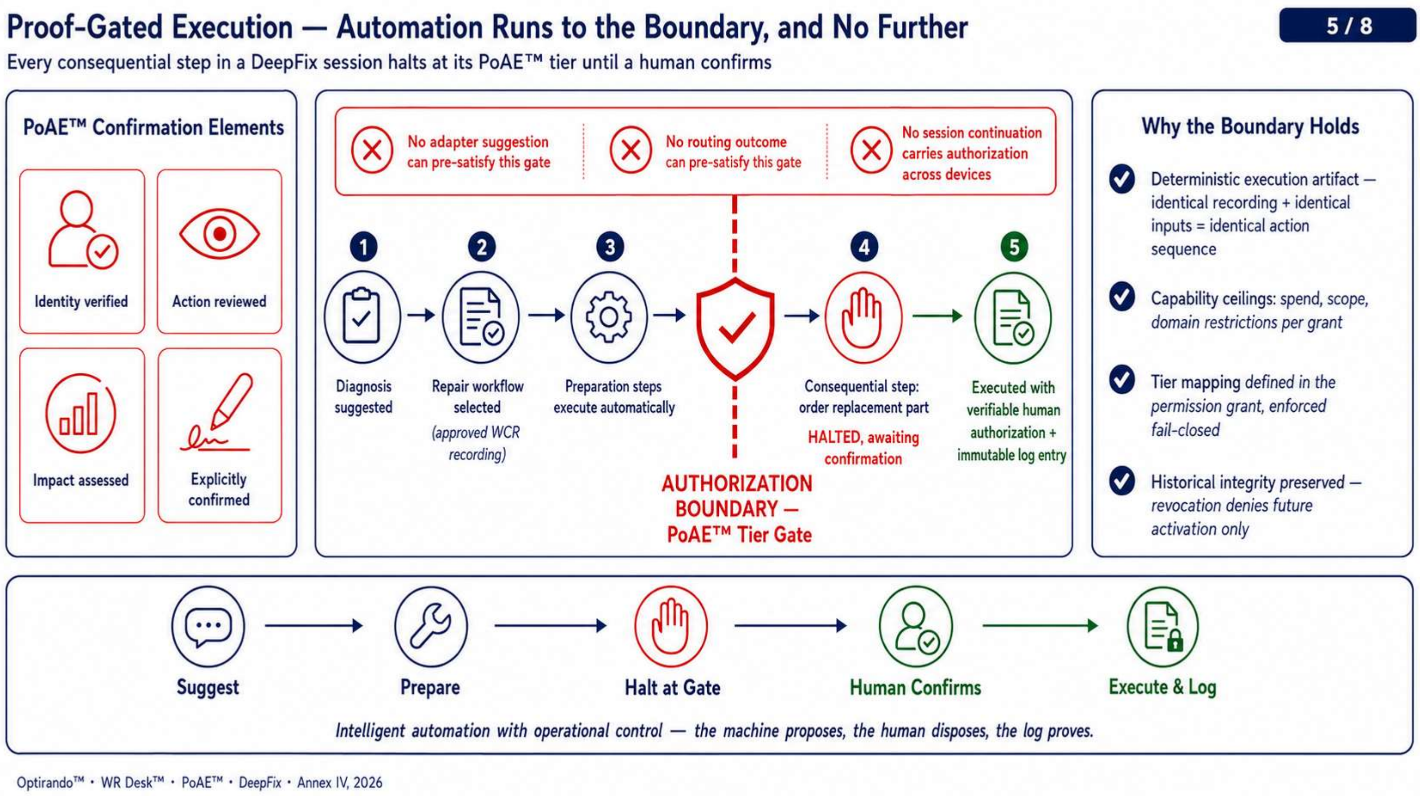
Proof-Gated Execution — Automation Runs to the Boundary, and No Further
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Every consequential step in a DeepFix session halts at its PoAE™ tier until a human confirms
PoAE™ Confirmation Elements
Identity verified
Action reviewed
Impact assessed
Explicitly confirmed
No adapter suggestion can pre-satisfy this gate
No routing outcome can pre-satisfy this gate
No session continuation carries authorization across devices
1
Diagnosis suggested
2
Repair workflow selected (approved WCR recording)
3
Preparation steps execute automatically
AUTHORIZATION BOUNDARY — PoAE™ Tier Gate
4
Consequential step: order replacement part
HALTED, awaiting confirmation
5
Executed with verifiable human authorization + immutable log entry
Why the Boundary Holds
Deterministic execution artifact — identical recording + identical inputs = identical action sequence
Capability ceilings: spend, scope, domain restrictions per grant
Tier mapping defined in the permission grant, enforced fail-closed
Historical integrity preserved — revocation denies future activation only
Suggest
Prepare
Halt at Gate
Human Confirms
Execute & Log
Intelligent automation with operational control — the machine proposes, the human disposes, the log proves.
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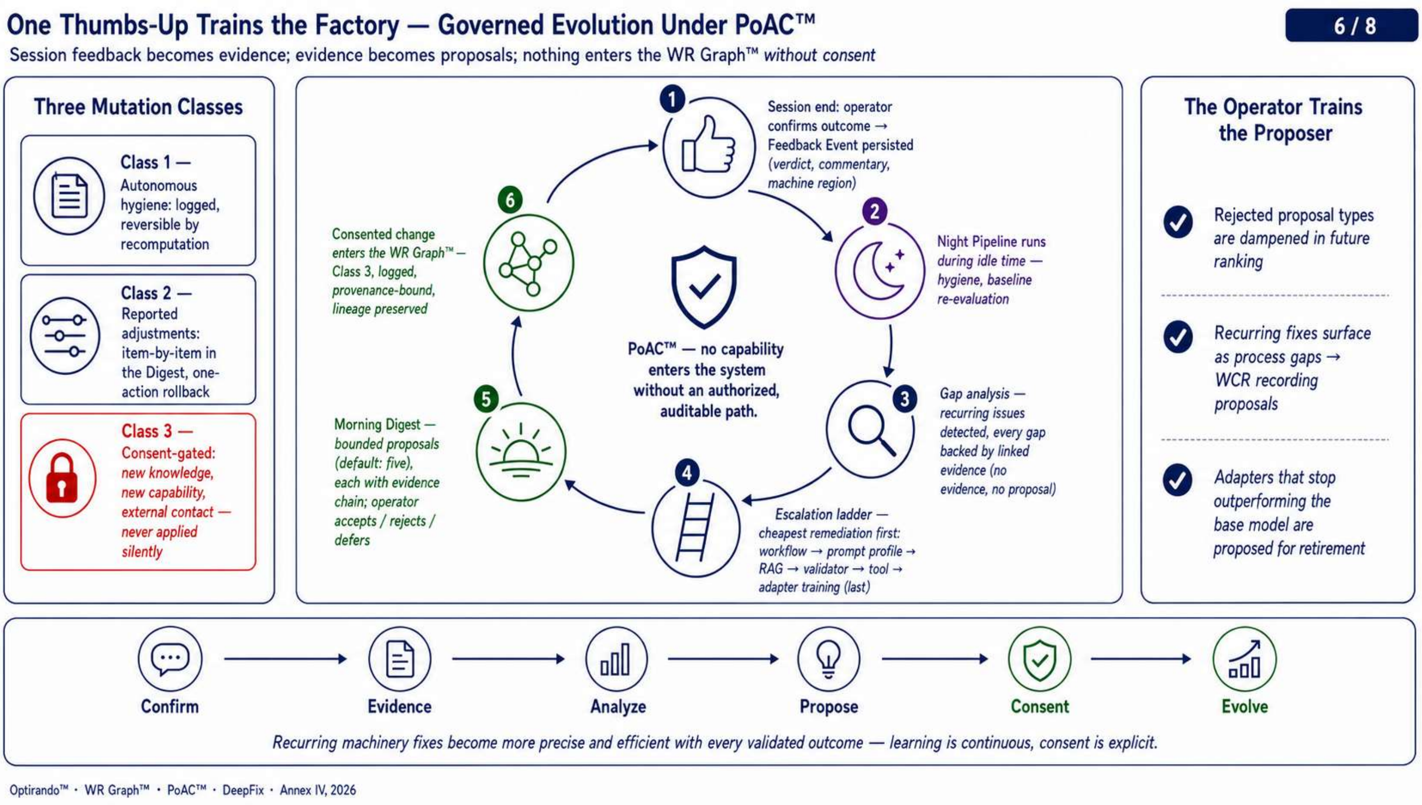

One Thumbs-Up Trains the Factory — Governed Evolution Under PoAC™
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Session feedback becomes evidence; evidence becomes proposals; nothing enters the WR Graph™ without consent
Three Mutation Classes
Class 1 — Autonomous hygiene: logged, reversible by recomputation
Class 2 — Reported adjustments: item-by-item in the Digest, one-action rollback
Class 3 — Consent-gated: new knowledge, new capability, external contact — never applied silently
1
Session end: operator confirms outcome → Feedback Event persisted (verdict, commentary, machine region)
2
Night Pipeline runs during idle time — hygiene, baseline re-evaluation
3
Gap analysis — recurring issues detected, every gap backed by linked evidence (no evidence, no proposal)
4
Escalation ladder — cheapest remediation first: workflow → prompt profile → RAG → validator → tool → adapter training (last)
5
Morning Digest — bounded proposals (default: five), each with evidence chain; operator accepts / rejects / defers
6
Consented change enters the WR Graph™ — Class 3, logged, provenance-bound, lineage preserved
PoAC™ — no capability enters the system without an authorized, auditable path.
The Operator Trains the Proposer
Rejected proposal types are dampened in future ranking
Recurring fixes surface as process gaps → WCR recording proposals
Adapters that stop outperforming the base model are proposed for retirement
Confirm
Evidence
Analyze
Propose
Consent
Evolve
Recurring machinery fixes become more precise and efficient with every validated outcome — learning is continuous, consent is explicit.
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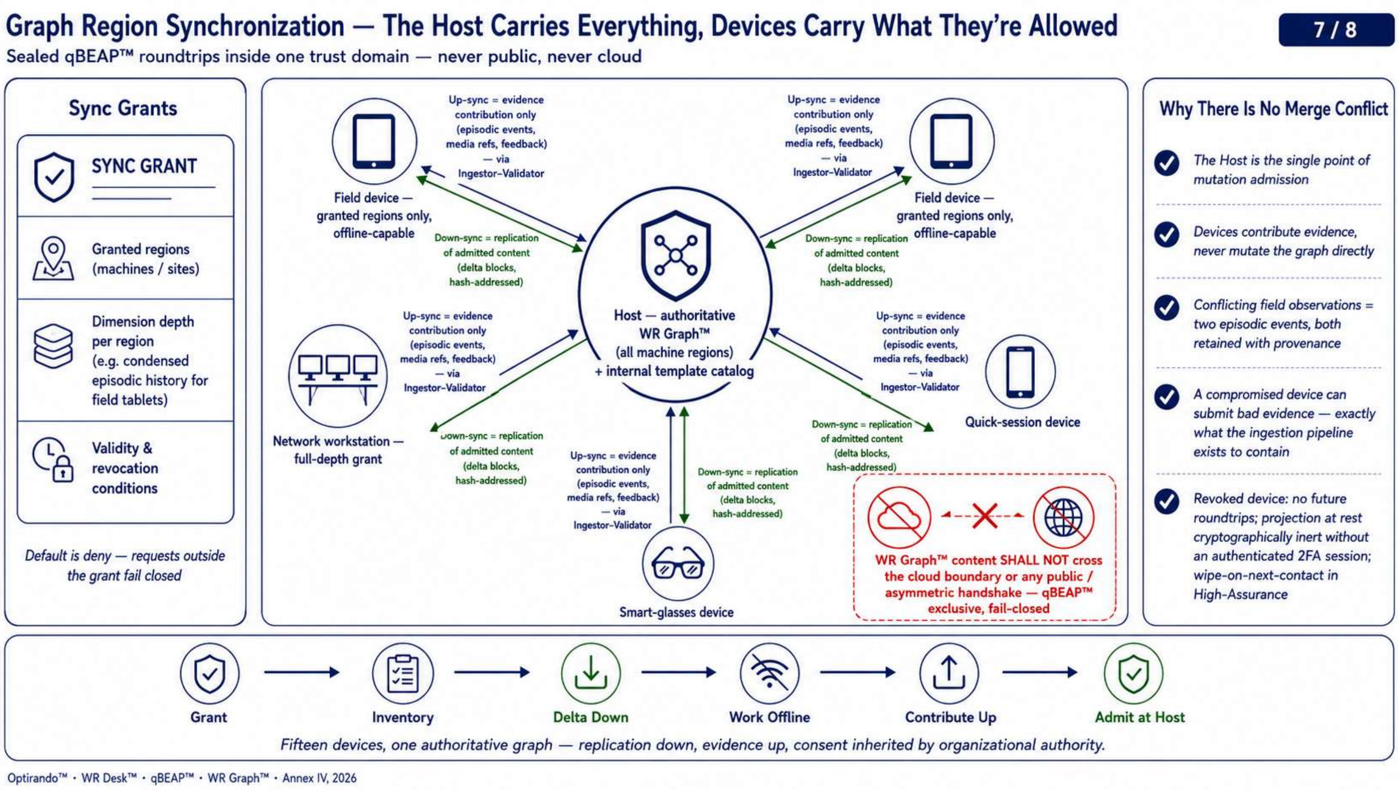

Graph Region Synchronization — The Host Carries Everything, Devices Carry What They're Allowed
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Sealed qBEAP™ roundtrips inside one trust domain — never public, never cloud
Sync Grants
SYNC GRANT
Granted regions (machines / sites)
Dimension depth per region (e.g. condensed episodic history for field tablets)
Validity & revocation conditions
Default is deny — requests outside the grant fail closed
Field device — granted regions only, offline-capable
Field device — granted regions only, offline-capable
Up-sync = evidence contribution only (episodic events, media refs, feedback) — via Ingestor-Validator
Down-sync = replication of admitted content (delta blocks, hash-addressed)
Host — authoritative WR Graph™ (all machine regions) + internal template catalog
Network workstation — full-depth grant
Quick-session device
Smart-glasses device
WR Graph™ content SHALL NOT cross the cloud boundary or any public / asymmetric handshake — qBEAP™ exclusive, fail-closed
Why There Is No Merge Conflict
The Host is the single point of mutation admission
Devices contribute evidence, never mutate the graph directly
Conflicting field observations = two episodic events, both retained with provenance
A compromised device can submit bad evidence — exactly what the ingestion pipeline exists to contain
Revoked device: no future roundtrips; projection at rest cryptographically inert without an authenticated 2FA session; wipe-on-next-contact in High-Assurance
Grant
Inventory
Delta Down
Work Offline
Contribute Up
Admit at Host
Fifteen devices, one authoritative graph — replication down, evidence up, consent inherited by organizational authority.
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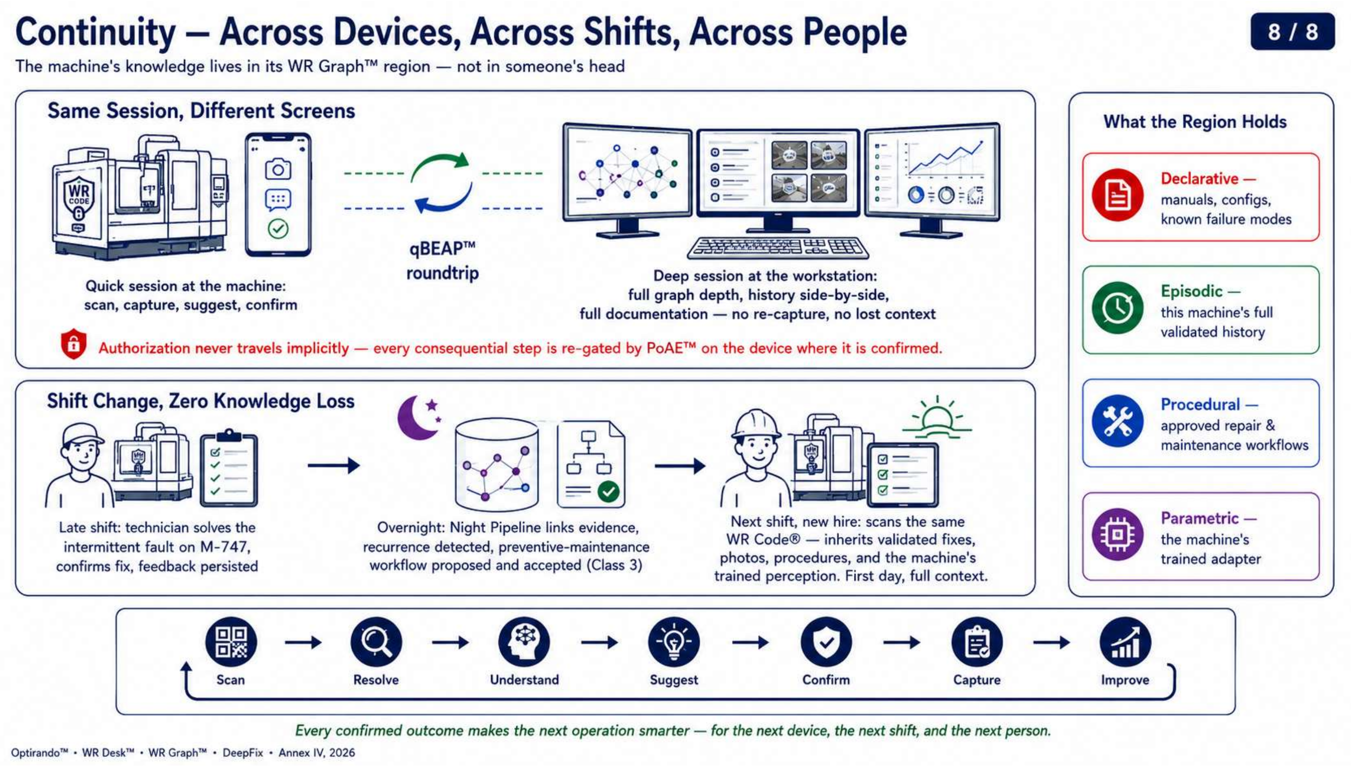
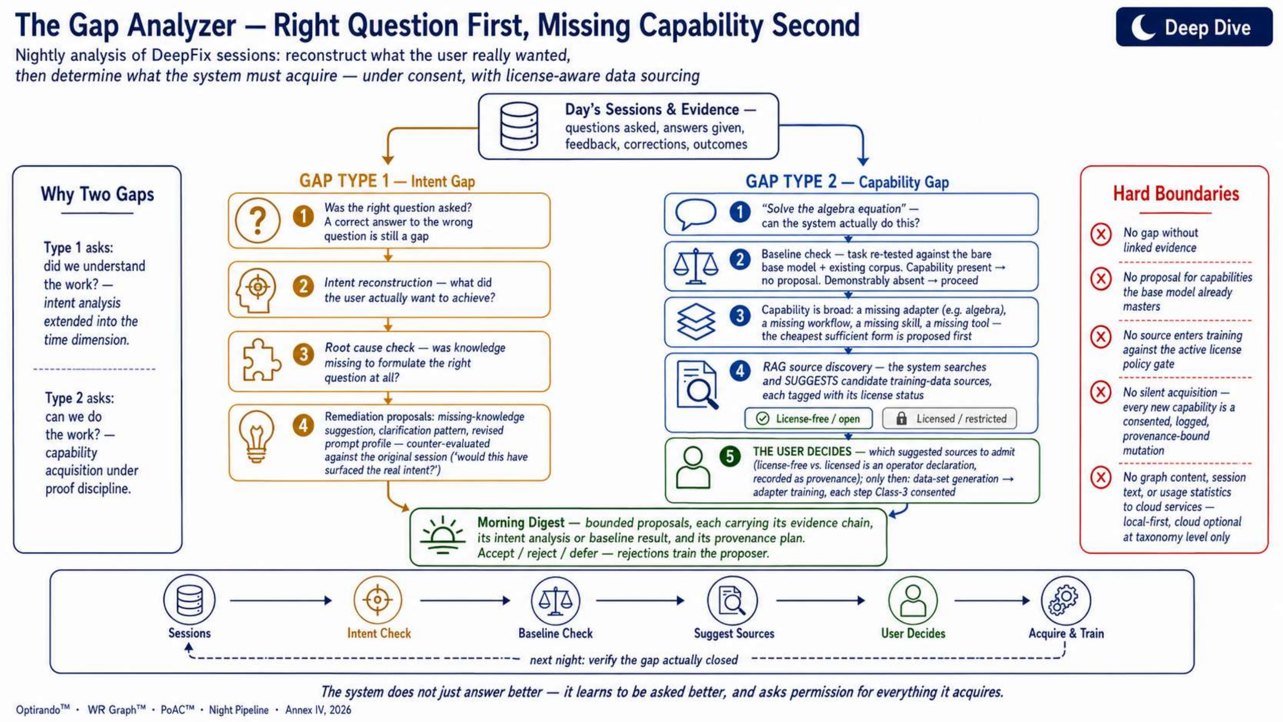


The Night Pipeline Gap Analyzer (Informative)

The preceding figures depict the DeepFix session as experienced at runtime: resolution, contextual suggestion, gated execution, evidence capture, synchronization, and continuity. The figure below depicts the complementary offline process. During idle periods, the gap analyzer of the Night Pipeline re-examines the day's sessions along two axes. The first is intent: whether the question posed corresponded to the operator's actual objective — a correct answer to a mis-posed question constitutes a gap, and remediation targets the knowledge required to pose the question correctly. The second is capability: whether the competence to perform the requested work exists at all, verified against the bare base model before any proposal is generated. Where a capability deficit is confirmed, remediation ascends a cost-ordered ladder — workflow, prompt profile, retrieval enrichment, validator, tool integration — with adapter training as the final rung; candidate training sources are discovered and presented with their declared license status, and admission of any source is an operator decision recorded as provenance. Autonomous overnight operation is confined to derived, reconstructible hygiene mutations; every acquisition of new knowledge or capability is deferred to an explicit, evidence-backed proposal in the Morning Digest.



End of Annex IV.